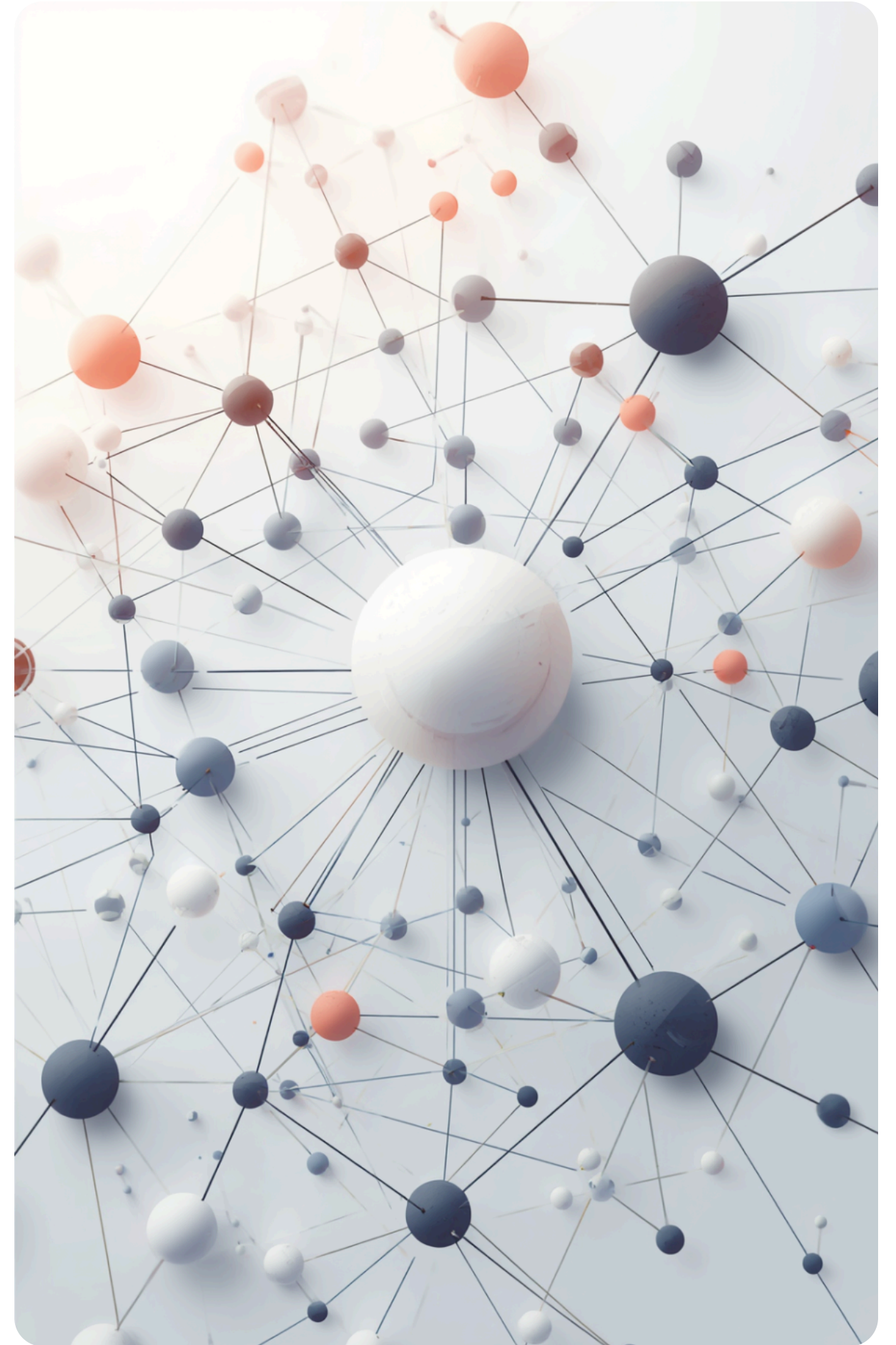
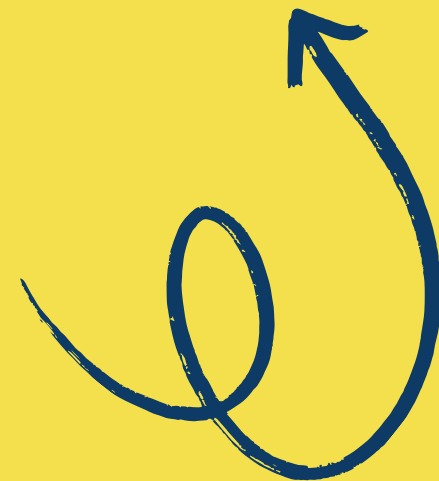


Customer Churn Prediction

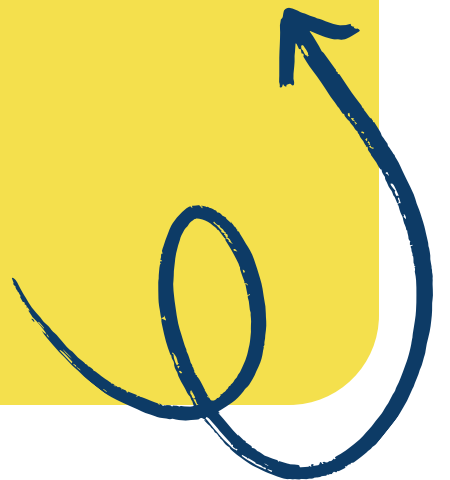
Ahmed Amin, MSC-KFS Data
Science Committee



Problem Statement

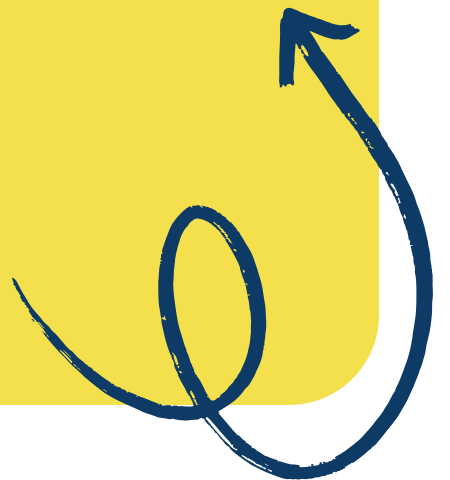
Predicting customer churn effectively

Churn occurs when customers stop using a service, directly impacting revenue and **customer lifetime value**. Our goal is to predict potential churn, enabling proactive retention strategies.



Overview of the dataset characteristics

The dataset comprises **505,206 customer records** that have been meticulously merged from training and testing sets. It contains **15 distinct features** that encompass demographic, behavioral, and subscription-related data. The target variable indicates customer **churn** in a binary format (0 for no churn, 1 for churn). The class distribution is notably balanced, with **55%** of customers representing churn and **45%** not churning. Furthermore, only **one null row** was dropped, ensuring data integrity and enhancing the reliability of our analysis. This dataset forms a solid foundation for predictive modeling efforts.



Exploratory Data Analysis

Key correlations and insights from data

Support Calls

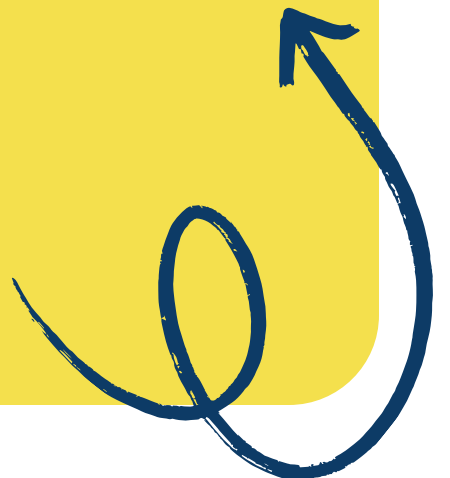
Support Calls were found to be **strongly correlated** with customer churn, indicating that increased inquiries often precede a customer's decision to leave the service.

Contract Length

Monthly customers exhibited significantly higher churn rates, suggesting that shorter contract lengths may contribute to a customer's likelihood of discontinuing their service.

Payment Delay

A higher delay in payments correlates with an increased churn rate, highlighting that timely payments are essential for maintaining customer relationships and minimizing churn.



Preprocessing Pipeline

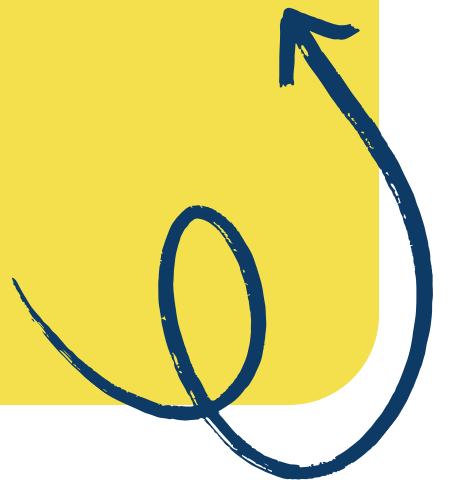
Efficient data preparation process

Data Merging

The training and testing datasets were merged to maximize the available data for training the model, ensuring a more robust performance during evaluation.

Feature Processing

Numerical features were scaled using StandardScaler, while categorical variables were transformed with OneHotEncoder. This preprocessing was executed within an sklearn Pipeline to prevent data leakage.



Model Comparison

Evaluating performance across algorithms

Logistic Regression

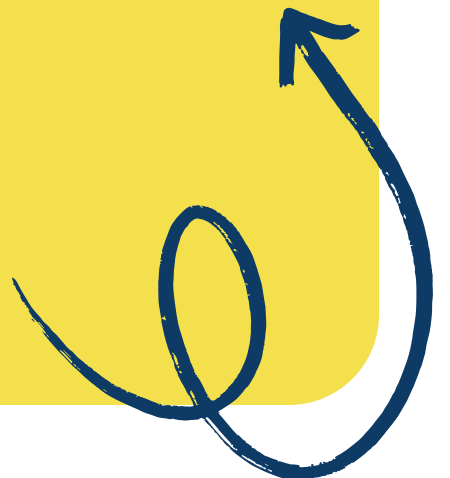
The baseline model achieved an accuracy of **0.840**, providing a fundamental starting point for evaluating more complex algorithms in churn prediction.

Gradient Boosting

This algorithm significantly improved performance with an accuracy of **0.913**, demonstrating its effectiveness in handling the nuances of customer data for churn prediction.

XGBoost

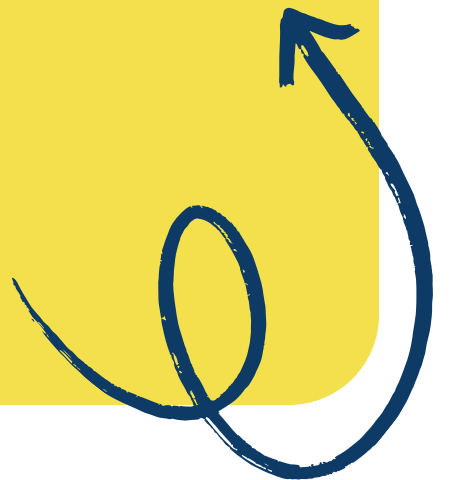
Achieving an accuracy of **0.933**, XGBoost showcased robust performance, leveraging advanced boosting techniques to enhance predictive capabilities for customer churn analysis.



Best Model

Tuned LightGBM Achieving 93.64% Accuracy

The tuned LightGBM model, with optimized parameters, achieved impressive cross-validated accuracy of **93.64%**, showcasing its effectiveness in predicting customer churn and enhancing retention strategies.



What Did Not Work

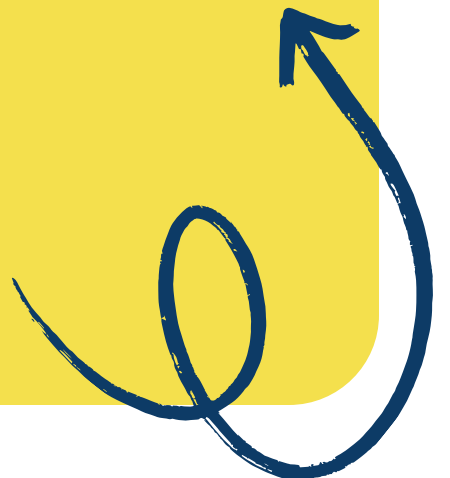
Limitations in feature engineering and stacking

Feature Engineering

The attempt at **feature engineering** did not yield improvements, as the model performance remained under baseline levels, indicating that tree models inherently capture interactions effectively.

Model Stacking

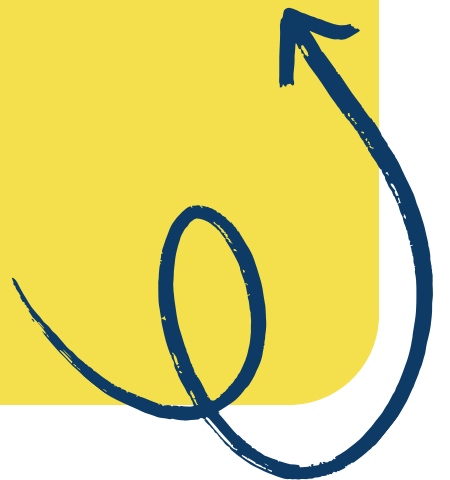
Stacking models like **LGBM** with Random Forest and Logistic Regression provided negligible gains, as they produced similar errors without additional predictive signals, highlighting the ceiling of this dataset's information.



Final Results

93.64% cross-validated accuracy achieved

The tuned LightGBM model demonstrated exceptional performance, achieving **93.64% cross-validated accuracy** on unseen data, validating its effectiveness for predicting customer churn and enhancing retention strategies.



Conclusion



Explore more at GitHub repository